CS 203 Assignment 1: AgentEvalOps Dataset (Project 1)

Kshitij Giri (23110177) · Shardul Junagade (23110297) · Vedant Sharan (23110354)

Repository: https://github.com/VedantSharan/AgentEvalOps (tag `a1-final`).

Every number carries an evidence ID such as [C3]. Appendix A maps each ID to the file and field that holds it, and `python -m code.reporting.check_report_numbers` re-reads all of them and confirms they match.

1. Collection and instrumentation (Part A)

A plain-Python function-calling agent uses five typed tools over a synthetic environment:

- `retrieve`: keyword search over a knowledge base;
- `calculate`: safe arithmetic;
- `query_records`: an employee table;
- `read_file`: virtual CSV and JSON files;
- `submit_answer`.

Task suite. 6 template families × 5 instances, 30 tasks in total [A1, A2]. Gold answers are computed by solver functions, and the suite includes 5 ambiguous or unanswerable tasks.

Collection and Tracing Architecture. A queue-based worker (`code/experiments/trajectory_worker.py`) ran the agent on 21 Oct 2026 through 5 hosted free-tier APIs and 10 models [A3–A5], at temperature 0 with at most 5 LLM turns. This produced **154 trajectories**. 52 runs were logged as `error`, and 42 ended without a submitted answer [A7, A8]. Each trace stores the task, configuration hash, model and prompt version, every LLM and tool call with arguments, outputs, errors, latency and tokens, and the final answer.

The telemetry system supports two complementary execution modes via MLflow Tracing (`code/tracing/mlflow_tracer.py`): 1. **Synchronous inline tracing (`code/experiments/runner.py`):** Wraps the agent's execution loop directly with `MLflowTracer` for single-agent evaluation benchmarks, generating hierarchically nested spans (agent execution root, LLM calls with token usage and latency, tool invocations with typed arguments/returns, and deterministic grading evaluation) live during execution. 2. **Decoupled batch ingestion (`code/tracing/export_trajectories_to_mlflow.py`):** In batch multi-provider runs across 10 hosted models, `trajectory_worker.py` logged rich trajectories to append-only JSONL files and a durable SQLite queue, intentionally decoupling live API execution from the MLflow SQLite backend (`mlruns/mlflow.db`). This architectural decision eliminated SQLite concurrency lock contention (`OperationalError: database is locked`) during concurrent multi-provider streaming and isolated external API rate-limit backoffs. The offline ingestion adapter faithfully maps the complete telemetry into MLflow traces and spans with zero fabrication. The resulting 174 traces and 633 spans across 43 tasks are inspectable via `mlflow ui`.

Coverage and Provenance:

- **Task coverage:** Trajectories cover 20 of the 30 tasks [A6], prioritizing multi-model depth with 6–10 runs per task across 10 hosted models (154 total). This satisfies all annotation, detector, and family-level split requirements, while leaving the remaining 10 tasks as an unexecuted benchmark reserve for Assignment 2.
- `data/metadata.csv` provides complete provenance across all 154 collected trajectories with authentic execution timestamps recovered from the work-queue database, the 30 base benchmark tasks, and all 149 augmented tasks (381 records total).

Data flow: `docs/DATA_COLLECTION.md`.

2. Data validation (Part B)

The 12-check suite (`code/validation/benchmark_validator.py`) covers JSON parsing, Pydantic schemas, unique IDs, reference integrity, content hashes, duplicates, answer leakage into prompts, tool determinism, tool-call arguments, metric ranges, missing values and grader consistency.

Before and after. Initially the suite failed on answer leakage in `T5_audit_04`; the prompt was fixed (`docs/VALIDATION_REPORT.md`). Saved evidence files:

Dataset	Checks passed	Checks failed
Raw (154 runs)	12/12 [B1]	0
Processed	12/12 [B2]	0
Deliberately corrupted fixtures	1/12	11 [B3]
Augmented tasks	12/12 [B4]	0

A finding from labelling. The logged grader (v1) rejected correctly formatted answers, because `float("$25,920")` fails, and it required the literal word "unanswerable". Grader v2 normalises currency, separators, units and sentences, and accepts explicit abstentions:

- correct answers rise from 48 to 87 [B5, B6];
- 39 answers flip from wrong to correct, and 0 the other way [B7, B8].

Its agreement with humans is reported out of sample in Section 3.

3. Labelling and agreement (Part C)

Label levels, filled in this order (`LABELLING_PROTOCOL.md` §4):

1. `outcome`
2. `primary_failure_category` (7 classes)
3. `first_error_step`

Rounds. Two annotators (A1, A2) labelled independently in separate Label Studio projects:

- **Round 1** (guidelines v1): 45 stratified trajectories [C1].
- **Round 2** (guidelines v2): 40 new trajectories [C2]. These were drawn with seed 2026 from the 109 unlabelled runs and deliberately enriched with failures: prevalence 0.625 vs 0.16 [C14, C15].

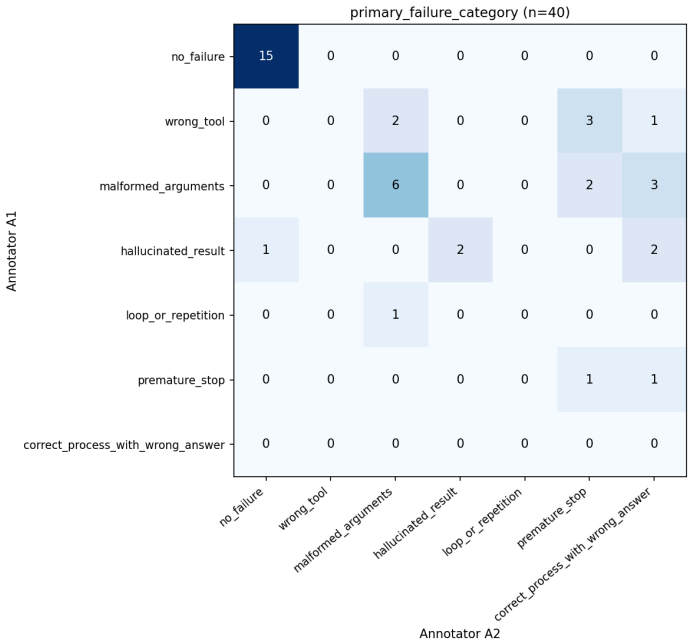
Both rounds overlap fully.

Adjudication. Disagreements (8 in round 1, 19 in round 2 [C18, C19]) were resolved by ADJ1, which is A2 acting as adjudicator:

- Each proposal was drafted with AI assistance from trace evidence. ADJ1 confirmed all 27 and changed 2 [C25].
- Agreed labels were never overridden.
- Result: 85 labelled trajectories, 33 of them failures, 27 adjudicated [C34–C36].

Table 1. Agreement before and after the guideline revision (Cohen's κ, scikit-learn; matched by `run_id`)

Measure	Round 1 (v1)	Round 2 (v2)
Outcome κ (PABAK)	0.85 (0.91) [C3, C12]	0.95 (0.95) [C4, C13]
Category κ, all items	0.45 [C5]	0.48 [C6]
Category κ, items both called failure	0.05 (n = 7) [C7]	0.21 (n = 24) [C8, C9]
First-error step κ (no failure coded −1)	0.65 [C10]	0.87 [C11]
Median seconds per item (A1)	4.2 [C16]	34.5 [C17]



Round-2 confusion matrix for `primary_failure_category` (rows: A1, columns: A2). Agreement concentrates on `no_failure` and `malformed_arguments`; A1's `wrong_tool` and A2's `correct_process_with_wrong_answer` do not overlap.

What changed in v2 (`docs/annotation_guidelines_v2.md`, changes table):

- `wrong_tool` restricted to tools that *cannot* supply the information;
- `malformed_arguments` = the tool rejected the arguments;
- a first-deviation rule for the step, left empty on success;
- a rule for valid but unproductive calls;
- format-insensitive correctness;
- the grader verdict hidden from annotators.

Results. Outcome and step agreement improved. Category agreement on genuine failures improved, but stays low.

Caveats:

- κ depends on prevalence, and round 2 has about 4× more failures, so we also show PABAK and the both-failure subset.
- A post-hoc 4-class merge of the taxonomy helps only slightly (0.28 [C37]). The disagreement is systematic between annotators: A1 used `wrong_tool` 6 times, A2 never; A2 used `correct_process_with_wrong_answer` 7 times, A1 never [C38–C40].

Disagreements. All 27 are discussed with trace evidence in Appendix C; three recurring patterns, plus one gap in agreed labels:

1. `wrong_tool` vs `malformed_arguments` when `query_records` is asked for fields it does not have.
2. "Step-limit wandering": the answer was found and never submitted.
3. Step disagreements where an annotator marked where the run *ended*, not where it went wrong.
4. **Gap in agreed labels:** three empty submissions on unanswerable tasks were labelled success by both annotators.

v3 recommendations: `docs/annotation_findings.md` §5.

Human vs grader. In round 1, the humans corrected 13 runs that grader v1 had logged as failures [C20]. On round 2, which grader v2 had never seen, agreement with humans is κ 0.83 and 0.78, against 0.52 and 0.48 for v1 [C21–C24].

Automatic detectors vs human primary labels (`supporting_files/detectors/`):

- **Malformed calls.** "The tool rejected the arguments" (file-not-found excluded) reaches precision 0.89, recall 0.80, κ 0.82 [C26–C28]. Against each annotator separately it scores κ 0.40 (A1) and 0.88 (A2) [C29, C30].
- **Loops.** "≥3 identical calls" finds only 0.25 of the human-labelled loops [C31]. Adding "step limit without an answer" finds all of them, at precision 0.29 [C32, C33]. Most false positives are real events that humans did not choose as the *primary* cause.

Gold test set: frozen in Part E.

4. Augmentation and active learning (Part D)

We built 149 augmented tasks [D1]. Every one comes with an executed check, not just code; all checks pass [D5–D7].

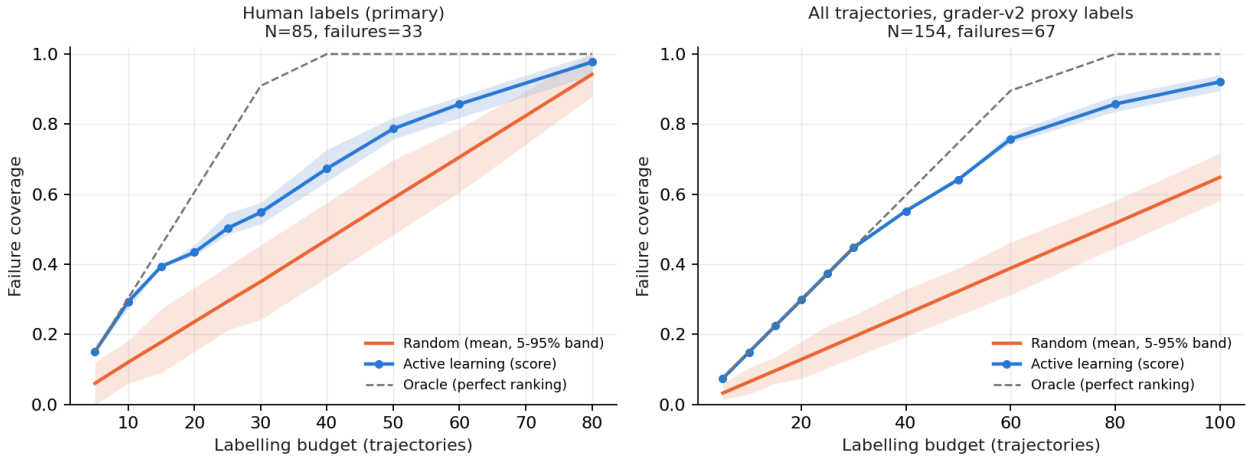
1. **Entity and parameter variants** (23) [D2]. Gold answers are recomputed by the original solver functions. Each variant's evidence is then confirmed reachable by calling the real tools. This check *rejected* 3 plausible candidates whose facts the tools cannot reach [D8].
2. **Paraphrases** (30) [D3]. Rule-based and seeded. A protected-span check (numbers, IDs, quoted names, acronyms, logic words) must match exactly, and negative tests show it catches dropped or added conditions. In a human spot check, 10/10 paraphrases kept their intent [D10].
3. **Fault injection** (96) [D4]. A wrapper around the real tool registry injects four faults: transient failure, outage, +2 s latency, and changed output format. Each fault is checked to behave as specified:
   - Format changes carry identical canonical content.
   - An **information outage** takes down all three information tools, so the expected behaviour *changes* to abstaining. 19 tasks are re-labelled unanswerable this way [D9].
   - A calculator outage keeps the gold answer, because the arithmetic can be done by hand.
   - Faults never mark themselves in tool output. Injected faults are identified by the task's fault spec and the injector log.

No agent runs were made on augmented tasks (no API keys).

Active learning vs random. Selection is by a label-free score: normalised step count + repeated identical calls + tool errors. It is compared with random selection on the same 85 human-labelled trajectories (33 failures [D23]), with 1,000 seeded trials [D24] and the same budgets and metric.

Budget	Failure coverage, active learning	Failure coverage, random (mean)
10	0.29 [D11]	0.12 [D12]
20	0.43 [D13]	0.24 [D14]
30	0.55 [D15]	0.35 [D16]

Random selection matched active learning in at most 0.1% of trials [D17].



Failure coverage vs labelling budget. Left: human labels (primary). Right: all 154 runs with grader-v2 proxy labels. Bands: 5th to 95th percentile over 1,000 seeds.

What the curve shows:

- The score finds failures faster, but it favours long, error-heavy runs. 13 of its top 30 never submitted an answer [D20], so random selection finds *more distinct categories* at budget 30 (0.79 vs 0.67 [D18, D19]).
- On the proxy pool the gap is larger (0.45 vs 0.19 at budget 30 [D21, D22]), but this is partly trivial: many picks are runs without an answer.
- The labelled pool is not a random sample, because round 2 was enriched with failures.

5. Versioning and splits (Part E)

Splitting unit: template family; assignment is deterministic. The repository's own plan put T1–T3 in train, T4 in validation and T5 in test. It called T6 "stratified", which would place one family in several splits, so the whole of T6 goes to test.

Split	Families	Trajectories
train	3 families	78 [E1]
validation	1 family	32 [E2]
test (gold)	2 families	44 trajectories, 24 human-labelled [E3, E4]
shifted	96 fault-injected tasks [E5]	evaluation only

What follows its parent's split: variants, repeated runs and labels.

Ten automatic checks pass [E6]. They test that families are disjoint, that no item and no identical prompt is in two splits, that each item has the right parent, that the shifted split equals the fault-injected tasks, that every manifest row's hash matches its source record, that every data file is either split or excluded with a reason, and the gold checksums. Negative tests confirm each check fails on a tampered manifest.

Gold-set SHA-256:

- `gold_tasks.jsonl` `f4b15d31…` [E7]
- `gold_labels.csv` `72e8f84f…` [E8]

DVC. Raw inputs are tracked with `dvc add`. A 14-stage `dvc.yaml` rebuilds every derived file, byte-identical in testing, and pushes to a local remote. Human inputs (raw exports, signed adjudications, spot check) stay in git. The repository is tagged `a1-final` (`HANDOFF_A2.md` §6).

6. Dataset card, software and labelling protocol (Part F)

Dataset card (`docs/DATASET_CARD.md`):

- 30 synthetic tasks;
- 154 trajectories from 10 hosted models;
- 85 human labels: 52 success and 33 failures, mostly correct-process-wrong-answer (11) and malformed arguments (10) [F1–F3];
- 149 augmented tasks.

Known biases:

- uneven model coverage (2 to 20 runs per model) [F4, F5];
- 10 tasks never run;
- non-random labelled samples;
- noisy categories;
- an adjudicator who is also an annotator;
- a keyword-only `retrieve` tool that shapes the observed failures.

Software (`ENVIRONMENT.md`):

- Python 3.10.12 on Ubuntu 22.04.5;
- DVC 3.67.1;
- Label Studio 1.23.2 (round 2; the round-1 version was not recorded).

All packages are pinned (`requirements.txt`), with the full freeze in `requirements_freeze.txt`. No GPU or local model was used. Model identity is the exact API model string; providers expose no revision hashes. The seeds are 42 (tasks, collection order, round-1 sample), 2026 (round-2 sample and order) and 203 (augmentation, active learning).

Labelling protocol (`LABELLING_PROTOCOL.md`): it records the anonymised IDs; the items per annotator and round; the config and guideline per round; the level order; the item order, which was sequential and followed in both rounds; what annotators saw (in round 1 they saw grader v1's verdict, removed in round 2); and the adjudication process. Export filenames, timestamps and checksums are listed per round.

7. Contribution table

Member	Roll Number	Primary Contributions
Vedant Sharan	23110354	Project architecture & data contracts; Part A benchmark generation (24 tasks, revised 30 tasks) & gold grading; tool registry & function-calling agent loop; Part B dataset validation suite (10 automated checks); persistent multi-provider collection pipeline & work queue; execution of 154 live trajectories across 10 models; stratified sampling for labelling & Label Studio XML config; round-1 and round-2 human annotation (Annotator A1); pipeline forensic audit & remediation; MLflow trace telemetry ingestion; Part A provenance metadata resolution (100% sample coverage across 154 runs, 30 tasks, 149 augmented tasks with work-queue timestamps).
Kshitij Giri	23110177	Round-2 human annotation (Annotator A2); adjudication of disagreements (ADJ1 across rounds 1 and 2); grader v2 implementation; paraphrase human spot check; Part C agreement analysis (kappa, confusion matrices, automatic detectors); Part D augmentation pipeline & active learning vs random experiments; Part E family-level splits & leakage verification suite; DVC pipeline setup (`dvc.yaml`, local remote migration); Part F reproducibility documentation (README, ENVIRONMENT, LABELLING_PROTOCOL, HANDOFF_A2, dataset card), packaging script, and initial report drafting.
Shardul Junagade	23110297	Project proposal drafting and research problem formulation; domain research on agent failure modes, evaluation taxonomy, and LLM provider quotas/pricing; review of annotation guidelines and edge cases; report review and verification of findings against submission requirements.

8. Limitations and open items

1. **Provenance metadata (Part A, resolved).** `data/metadata.csv` now provides complete provenance across all 154 runs with authentic microsecond-level timestamps recovered from the SQLite work queue, alongside the 30 base tasks and 149 augmented tasks.
2. **Size.** 154 trajectories, below the brief's "a few hundred", and the augmented tasks have no agent runs.
3. **Fast round-1 annotation.** About 5 s per item suggests shallow reading. Round 2 took about 30 s per item.
4. **Low category reliability.** This is the main data-quality risk for Assignment 2.
5. **Trace telemetry integration.** Synchronous MLflow tracing is integrated in `code/experiments/runner.py`, and the batch multi-provider trajectories are ingested into `mlruns/mlflow.db` (174 traces, 633 spans across 43 tasks) via `code/tracing/export_trajectories_to_mlflow.py`, bridging telemetry into the MLflow UI for Assignment 2.

Appendix A: Evidence index

Every number in the report, the file that holds it, and how it is located. `python -m code.reporting.check_report_numbers` re-reads them all; the result is in `supporting_files/report_traceability.csv`.

ID	Claim	Value	File	Locator
A1	Base tasks	30	supporting_files/dataset_stats.json	json:tasks.n
A2	Template families (tasks per family = 5)	5	supporting_files/dataset_stats.json	json:tasks.by_family.multi_hop_retrieval
A3	Collected trajectories	154	supporting_files/dataset_stats.json	json:trajectories.n
A4	Distinct models	10	supporting_files/dataset_stats.json	json:trajectories.n_models
A5	Providers	5	supporting_files/dataset_stats.json	json:trajectories.n_providers
A6	Tasks with at least one run	20	supporting_files/dataset_stats.json	json:tasks.tasks_with_at_least_one_trajectory
A7	Runs with no submitted answer	42	supporting_files/dataset_stats.json	json:trajectories.no_submitted_answer
A8	Logged outcome error (grader v1)	52	supporting_files/dataset_stats.json	json:trajectories.logged_outcome_grader_v1.err
B1	Raw data: checks passed	12	supporting_files/validation/raw.json	json:checks_passed
B2	Processed data: checks passed	12	supporting_files/validation/processed.json	json:checks_passed
B3	Corrupted fixtures: checks failed	11	supporting_files/validation/corrupted.json	json:checks_failed
B4	Augmented tasks: checks passed	12	supporting_files/validation/augmented.json	json:checks_passed
B5	Answers correct under grader v1	48	supporting_files/grader_v2/regrade_summary.json	json:correct_v1
B6	Answers correct under grader v2	87	supporting_files/grader_v2/regrade_summary.json	json:correct_v2
B7	Answers v1 rejected but v2 accepts	39	supporting_files/grader_v2/regrade_summary.json	json:changed_false_to_true
B8	Answers v1 accepted but v2 rejects	0	supporting_files/grader_v2/regrade_summary.json	json:changed_true_to_false
C1	Round-1 items (full overlap)	45	supporting_files/agreement/pre_post_comparison.csv	csv:round=round1_v1:n_items
C2	Round-2 items (full overlap)	40	supporting_files/agreement/pre_post_comparison.csv	csv:round=round2_v2:n_items
C3	Round-1 outcome kappa	0.85	supporting_files/agreement/pre_post_comparison.csv	csv:round=round1_v1:outcome_kappa
C4	Round-2 outcome kappa	0.95	supporting_files/agreement/pre_post_comparison.csv	csv:round=round2_v2:outcome_kappa
C5	Round-1 category kappa	0.45	supporting_files/agreement/pre_post_comparison.csv	csv:round=round1_v1:category_kappa
C6	Round-2 category kappa	0.48	supporting_files/agreement/pre_post_comparison.csv	csv:round=round2_v2:category_kappa
C7	Round-1 category kappa, both-failure items	0.05	supporting_files/agreement/pre_post_comparison.csv	csv:round=round1_v1:category_both_failure_kapp
C8	Round-2 category kappa, both-failure items	0.21	supporting_files/agreement/pre_post_comparison.csv	csv:round=round2_v2:category_both_failure_kapp
C9	Round-2 both-failure items	24	supporting_files/agreement/pre_post_comparison.csv	csv:round=round2_v2:category_both_failure_n
C10	Round-1 step kappa (no_failure = -1)	0.65	supporting_files/agreement/pre_post_comparison.csv	csv:round=round1_v1:step_kappa_nominal
C11	Round-2 step kappa (no_failure = -1)	0.87	supporting_files/agreement/pre_post_comparison.csv	csv:round=round2_v2:step_kappa_nominal
C12	Round-1 outcome PABAK	0.91	supporting_files/agreement/pre_post_comparison.csv	csv:round=round1_v1:outcome_pabak
C13	Round-2 outcome PABAK	0.95	supporting_files/agreement/pre_post_comparison.csv	csv:round=round2_v2:outcome_pabak
C14	Round-1 failure prevalence A1	0.16	supporting_files/agreement/pre_post_comparison.csv	csv:round=round1_v1:failure_prevalence_A1
C15	Round-2 failure prevalence A1	0.625	supporting_files/agreement/pre_post_comparison.csv	csv:round=round2_v2:failure_prevalence_A1
C16	Round-1 median seconds per item A1	4.2	supporting_files/agreement/pre_post_comparison.csv	csv:round=round1_v1:median_seconds_per_item_A1
C17	Round-2 median seconds per item A1	34.5	supporting_files/agreement/pre_post_comparison.csv	csv:round=round2_v2:median_seconds_per_item_A1
C18	Round-1 items with any disagreement	8	supporting_files/agreement/round1/summary.json	json:n_items_with_any_disagreement
C19	Round-2 items with any disagreement	19	supporting_files/agreement/round2/summary.json	json:n_items_with_any_disagreement
C20	Round-1 logged failure, both humans success	13	supporting_files/agreement/round1/summary.json	json:human_vs_logged_outcome.logged_failure_bo
C21	Round-2 A1 vs grader v2 kappa (out of sample)	0.83	supporting_files/agreement/round2/human_vs_grader.summary.json	json:A1_vs_grader_v2.kappa
C22	Round-2 A2 vs grader v2 kappa (out of sample)	0.78	supporting_files/agreement/round2/human_vs_grader.summary.json	json:A2_vs_grader_v2.kappa
C23	Round-2 A1 vs grader v1 kappa	0.52	supporting_files/agreement/round2/human_vs_grader.summary.json	json:A1_vs_grader_v1.kappa
C24	Round-2 A2 vs grader v1 kappa	0.48	supporting_files/agreement/round2/human_vs_grader.summary.json	json:A2_vs_grader_v1.kappa
C25	Round-2 adjudications confirmed by ADJ1	19	supporting_files/agreement/round2/adjudication_round2.csv	csvcount:confirmed_by=ADJ1
C26	Detector malformed_strict precision (final labels)	0.89	supporting_files/detectors/detector_metrics.csv	csv:detector=malformed_strict;reference=final;
C27	Detector malformed_strict recall	0.80	supporting_files/detectors/detector_metrics.csv	csv:detector=malformed_strict;reference=final;
C28	Detector malformed_strict kappa	0.82	supporting_files/detectors/detector_metrics.csv	csv:detector=malformed_strict;reference=final;
C29	Detector malformed_strict kappa vs A1	0.40	supporting_files/detectors/detector_metrics.csv	csv:detector=malformed_strict;reference=A1;sub
C30	Detector malformed_strict kappa vs A2	0.88	supporting_files/detectors/detector_metrics.csv	csv:detector=malformed_strict;reference=A2;sub
C31	Detector loop_strict recall	0.25	supporting_files/detectors/detector_metrics.csv	csv:detector=loop_strict;reference=final;subse
C32	Detector loop_extended recall	1.0	supporting_files/detectors/detector_metrics.csv	csv:detector=loop_extended;reference=final;sub
C33	Detector loop_extended precision	0.29	supporting_files/detectors/detector_metrics.csv	csv:detector=loop_extended;reference=final;sub
C34	Human-labelled trajectories	85	supporting_files/dataset_stats.json	json:human_labels.n
C35	Human-labelled failures	33	supporting_files/dataset_stats.json	json:human_labels.outcome.failure
C36	Adjudicated labels	27	supporting_files/dataset_stats.json	json:human_labels.label_source.adjudicated
C37	Round-2 coarse 4-class kappa, both-failure (exploratory)	0.28	supporting_files/agreement/pre_post_comparison.csv	csv:round=round2_v2:coarse4_both_failure_kappa
D1	Augmented tasks	149	supporting_files/augmentation/summary.json	json:augmented_tasks
D2	Entity/parameter variants	23	supporting_files/augmentation/summary.json	json:by_type.entity_variant.tasks
D3	Paraphrases	30	supporting_files/augmentation/summary.json	json:by_type.paraphrase.tasks
D4	Fault-injected tasks	96	supporting_files/augmentation/summary.json	json:by_type.fault_injection.tasks
D5	Entity-variant checks failed	0	supporting_files/augmentation/summary.json	json:by_type.entity_variant.checks_failed
D6	Paraphrase checks failed	0	supporting_files/augmentation/summary.json	json:by_type.paraphrase.checks_failed
D7	Fault-injection checks failed	0	supporting_files/augmentation/summary.json	json:by_type.fault_injection.checks_failed
D8	Rejected variant candidates (evidence unreachable)	3	supporting_files/augmentation/summary.json	json:rejected_candidates

ID	Claim	Value	File	Locator
D9	Fault tasks whose expected behaviour changes (information outage)	19	`supporting_files/augmentation/summary.json`	`json:fault_label_effects.expected_behaviour_ch`
D10	Paraphrases judged intent-preserving by human	10	`supporting_files/augmentation/paraphrase_spot_check.csv`	`csvcount:intent_preserved (yes/no)=yes`
D11	AL failure coverage, human pool, budget 10	0.29	`supporting_files/active_learning/summary.json`	`json:pools.human.headline.10.active_learning`
D12	Random failure coverage, budget 10	0.12	`supporting_files/active_learning/summary.json`	`json:pools.human.headline.10.random_mean`
D13	AL failure coverage, budget 20	0.43	`supporting_files/active_learning/summary.json`	`json:pools.human.headline.20.active_learning`
D14	Random failure coverage, budget 20	0.24	`supporting_files/active_learning/summary.json`	`json:pools.human.headline.20.random_mean`
D15	AL failure coverage, budget 30	0.55	`supporting_files/active_learning/summary.json`	`json:pools.human.headline.30.active_learning`
D16	Random failure coverage, budget 30	0.35	`supporting_files/active_learning/summary.json`	`json:pools.human.headline.30.random_mean`
D17	P(random >= AL), budget 30	0.001	`supporting_files/active_learning/summary.json`	`json:pools.human.headline.30.p_random_ge_al`
D18	AL category coverage, budget 30	0.67	`supporting_files/active_learning/summary.json`	`json:pools.human.headline.30.al_category_cover`
D19	Random category coverage, budget 30	0.79	`supporting_files/active_learning/summary.json`	`json:pools.human.headline.30.random_category_c`
D20	AL top-30 picks with no submitted answer	13	`supporting_files/active_learning/summary.json`	`json:pools.human.headline.30.al_seed0_selected`
D21	Proxy pool AL coverage, budget 30	0.45	`supporting_files/active_learning/summary.json`	`json:pools.proxy_full.headline.30.active_learn`
D22	Proxy pool random coverage, budget 30	0.19	`supporting_files/active_learning/summary.json`	`json:pools.proxy_full.headline.30.random_mean`
D23	Human pool failures	33	`supporting_files/active_learning/summary.json`	`json:pools.human.failures`
E1	Train trajectories	78	`supporting_files/splits/leakage_report.json`	`json:counts.train.trajectory`
E2	Validation trajectories	32	`supporting_files/splits/leakage_report.json`	`json:counts.validation.trajectory`
E3	Test trajectories	44	`supporting_files/splits/leakage_report.json`	`json:counts.test.trajectory`
E4	Test human-labelled trajectories	24	`supporting_files/splits/leakage_report.json`	`json:counts.test.human_labelled_trajectories`
E5	Shifted (fault-injected) tasks	96	`supporting_files/splits/leakage_report.json`	`json:counts.shifted.augmented:fault_injection`
E6	All leakage checks passed	True	`supporting_files/splits/leakage_report.json`	`json:all_passed`
E7	Gold tasks SHA-256	f4b15d31aafd6ce4…	`data/splits/gold_test/gold_tasks.jsonl`	`sha256:`
E8	Gold labels SHA-256	72e8f84fa8363e0f…	`data/splits/gold_test/gold_labels.csv`	`sha256:`
C38	Round-2 A1 uses of wrong_tool (both-failure items)	6	`supporting_files/agreement/round2/summary.json`	`json:category_given_both_failure.distribution_`
C39	Round-2 A2 uses of correct_process_with_wrong_answer (both-failure items)	7	`supporting_files/agreement/round2/summary.json`	`json:category_given_both_failure.distribution_`
C40	Round-2 A2 uses of wrong_tool (both-failure items)	0	`supporting_files/agreement/round2/summary.json`	`json:category_given_both_failure.distribution_`
D24	Random/tie-break trials	1000	`supporting_files/active_learning/summary.json`	`json:trials`
F1	Human labels: success	52	`supporting_files/dataset_stats.json`	`json:human_labels.outcome.success`
F2	Human labels: correct_process_with_wrong_answer	11	`supporting_files/dataset_stats.json`	`json:human_labels.primary_failure_category.cor`
F3	Human labels: malformed_arguments	10	`supporting_files/dataset_stats.json`	`json:human_labels.primary_failure_category.mal`
F4	Fewest runs for one model (gemini-3.5-flash)	2	`supporting_files/dataset_stats.json`	`json:trajectories|by_model|gemini:gemini-3.5-f`
F5	Most runs for one model (e.g. cohere command-r7b)	20	`supporting_files/dataset_stats.json`	`json:trajectories.by_model.cohere:command-r7b-`

Appendix B: Use of AI assistance

Assistant. Claude (claude.ai agentic workspace, configured model `claude-opus-5-5`) was used on 22 Oct 2026 for Parts C (analysis), D, E and F. The exact prompts and the files each one produced are in `prompts/ai_assistance/claude_session_2026-10-22.md`. The assistant read the repository, wrote and ran code in a sandbox copy, and returned stage zips. The team unzipped them, reran every script on their own machine and committed the results.

What the assistant did:

- wrote the analysis, augmentation, active-learning, split, DVC and reporting code, with tests;
- drafted adjudication proposals from trace evidence;
- drafted guidelines v2 and these documents.

What humans did:

- every annotation (rounds 1 and 2);
- confirmation or change of every adjudication (ADJ1 changed 2 of 19 round-2 proposals);
- the paraphrase spot check;
- every command run and commit.

Prompts sent to models by the code are in `prompts/model_prompts/` and are verified against the code by `tests/test_prompts.py`. Any other assistant the team used must be listed in `prompts/ai_assistance/OTHER_TOOLS.md`.

Appendix C: Disagreements reviewed

All 27 items where the two annotators disagreed on at least one level. Shown for each: the A1 and A2 labels (outcome / category / step), the evidence from the trace, and the final decision. Full justifications and rule references: `supporting_files/agreement/round*/adjudication_round*.csv`.

Round	Run	Task	A1	A2	Trace evidence (abridged)	Final (ADJ1)
1	08eeee21	T3_struct_05	failure / wrong_tool / 1	failure / correct_process_with_wrong_answer / 1	Step 0 read_file(support_tickets.csv) succeeded and is the task's expected tool. Billing rows with a non-blank score != 5 are TCK-01/03/06/08/12 = 5 (= gold). Step 1 calculate('3 + 1 + 1 + 1 + 2') = 8 matches no valid count; step 2 submits 4, contradicting …	failure / correct_process_with_wrong_answer / 1
1	097b92f6	T6_unans_05	success / no_failure / 0	failure / hallucinated_result / 1	Task is unanswerable (empty log). Step 0 read_file returned '[]'. Step 1 LLM text says the average 'cannot be computed (or is effectively 0)' and ends 'Answer: 0'; step 2 submits '0'. Grader v2: failure (bare value).	failure / hallucinated_result / 1
1	0c9eefc9	T4_chain_04	failure / wrong_tool / 1	failure / premature_stop / 0	Expected tools: retrieve + calculate; the needed facts (subsidy %, base budget) exist in the knowledge base. retrieve only matches when every query word appears in a fact, so steps 0-2 (three different whole-sentence queries) each return 0 matches. Step 3-4…	failure / premature_stop / 0
1	26ff8938	T4_chain_02	failure / malformed_arguments / 2	failure / correct_process_with_wrong_answer / 2	Step 0 retrieve -> 'tax rate for Enterprise SaaS is 18.0%'. Step 1 read_file(q1_sales.csv) -> Enterprise SaaS revenue 20000 + 25000 = 45000. Step 2 calculate({'expression': '45000 * 0.82'}) = 36900, i.e. tax subtracted instead of added (gold 45000 x 1.18 = …	failure / correct_process_with_wrong_answer / 2
1	2ce94e88	T3_struct_03	failure / wrong_tool / 0	failure / correct_process_with_wrong_answer / 1	Expected tool: read_file. Step 0 read_file(server_metrics.csv) returned the full table; US-EAST + ONLINE + memory_usage_pct >= 70 gives 5 servers (= gold, recomputed). Step 1 submits 6 with no intermediate computation.	failure / correct_process_with_wrong_answer / 1
1	31eabcb1	T1_multihop_02	failure / wrong_tool / 0	failure / loop_or_repetition / 1	Expected tool: retrieve. Steps 0, 1 and 2 are the identical call retrieve(query=<same 25-word sentence>), each returning 0 matches. Step 3-4: agent gives up ('unable to find').	failure / loop_or_repetition / 0
1	4393465c	T5_audit_03	failure / wrong_tool / 2	failure / correct_process_with_wrong_answer / 2	Steps 0-1 read_file(sla_reference.json) -> PaymentGateway threshold 350, and read_file(logs/payment_api.json). Requests with latency_ms >= 350 for PaymentGateway = 5 (= gold, recomputed). Step 2 LLM outputs '7' with no computation; step 3 submits 7.	failure / correct_process_with_wrong_answer / 2
1	b77bee95	T6_unans_03	success / no_failure / 0	failure / hallucinated_result / 3	Ambiguous task (two employees named Rahul Sharma). Step 1 retrieve -> fact rf20: 'multiple employees named Rahul Sharma: Engineering (Systems Architect) and Sales (Regional Account Executive)'. Final answer reports both and does not pick one. Grader v2: cor…	success / no_failure / –
2	04dd387d	T4_chain_05	failure / wrong_tool / 4	failure / correct_process_with_wrong_answer / 4	Step 1 read_file('revenue_data.csv') not found; recovered at step 2 (consulting_ledger.csv: 18000+22000=40000); step 3 retrieve -> consulting tax 10%. Step 4 submits $36,000 = 40000 x 0.9; gold 44000 = 40000 x 1.1.	failure / correct_process_with_wrong_answer / 4
2	0ca1febf	T1_multihop_02	failure / premature_stop / 11	failure / premature_stop / 1	Step 0 retrieve returns rf21 and rf22; rf22 contains the superseded 2024 policy and the correct 180 USD value. Steps 1-10 search unrelated files; step 11 submits an empty answer.	failure / premature_stop / 1
2	1a4fe5f2	T4_chain_02	failure / malformed_arguments / 1	failure / premature_stop / 1	Step 0 read_file succeeds. Step 1 retrieve('official sales tax policy rate for Enterprise SaaS category') returns zero matches. The agent gives up and reports only the pre-tax total.	failure / premature_stop / 1
2	6b41cf74	T3_struct_05	failure / wrong_tool / 1	failure / premature_stop / 1	Step 0 reads support_tickets.csv, which contains the evidence needed for the count. Steps 1-8 include empty retrievals, a nonexistent file, unrelated reads, and an irrelevant query_records call; step 9 submits an empty answer.	failure / loop_or_repetition / 1
2	77461287	T1_multihop_03	failure / wrong_tool / 0	failure / malformed_arguments / 0	Steps 0-2 call query_records with fields project_name, project_manager, and on-call_escalation_sla; each call is rejected with an Unknown field error. The agent then says it cannot find the SLA.	failure / malformed_arguments / 0
2	85bce435	T3_struct_03	failure / hallucinated_result / 1	failure / correct_process_with_wrong_answer / 1	Step 0 reads server_metrics.csv. At step 1 the agent says there are 6 qualifying servers but lists the 5 correct servers: SRV-101, SRV-105, SRV-107, SRV-111, and SRV-113.	failure / correct_process_with_wrong_answer / 1
2	8e459d49	T3_struct_04	failure / malformed_arguments / 1	failure / correct_process_with_wrong_answer / 1	Step 0 reads support_tickets.csv and shows Billing tickets. Step 1 query_records filters department=Billing and aggregates maximum resolution_hrs, but the employee-table query returns zero rows. The agent concludes there are no Billing tickets.	failure / wrong_tool / 1
2	8e531d00	T4_chain_02	failure / malformed_arguments / 1	failure / premature_stop / 1	Step 0 read_file succeeds. Step 1 retrieve('official sales tax policy rate for Enterprise SaaS') returns zero matches. Steps 2-4 state that the rate cannot be found and stop.	failure / premature_stop / 1
2	961e13f8	T3_struct_05	failure / wrong_tool / 1	failure / malformed_arguments / 1	Step 0 reads support_tickets.csv. Steps 1-2 query_records with satisfaction_score are rejected as Unknown field. Steps 3-5 make additional ineffective retrieval calls; step 6 submits an empty answer.	failure / malformed_arguments / 1
2	a0a088d0	T4_chain_01	failure / hallucinated_result / 2	success / no_failure / –	Steps 0-2 obtain salary 95000, bonus rate 14.5%, and calculator output 13774.999999999998. Step 3 submits 13774.99; the gold answer is 13775.00 and the stated numeric tolerance is 0.005.	failure / correct_process_with_wrong_answer / 3
2	a31c4b99	T4_chain_04	failure / wrong_tool / 0	failure / premature_stop / 0	Step 0 read_file('corporate_infrastructure_policy.json')	failure / malformed_arguments / 0

Round	Run	Task	A1	A2	Trace evidence (abridged)	Final (ADJ1)
					returns file not found. The required subsidy rate is available in the knowledge base. The agent gives up at steps 1-2.	
2	a97af026	T4_chain_02	failure / premature_stop / 12	failure / correct_process_with_wrong_answer / 2	Steps 0-1 obtain revenue of 45000 and the correct 18% tax rate. Steps 2-11 contain repeated unproductive retrievals, nonexistent-file attempts, and unrelated reads. Step 12 submits an empty answer.	failure / loop_or_repetition / 2
2	b62ab151	T1_multihop_03	failure / malformed_arguments / 1	failure / malformed_arguments / 2	Step 0 read_file('projects.csv') returns file not found. Step 1 read_file('escalation_policies.csv') also returns file not found. The agent then gives up.	failure / malformed_arguments / 0
2	d1cbda85	T6_unans_02	success / no_failure / 2	success / no_failure / –	Both annotators labelled this unanswerable task as success/no_failure because the agent abstained rather than committing to a value. For a successful trajectory, first_error_step should be blank under R4.	success / no_failure / –
2	d4490fcf	T4_chain_02	failure / malformed_arguments / 3	failure / correct_process_with_wrong_answer / 3	Step 0 retrieval returns no matches; step 1 reads sales data; step 2 retrieves the correct 18% tax rate; step 3 calculate('45000 * 0.82') returns 36900, while the gold answer is 53100.	failure / correct_process_with_wrong_answer / 3
2	d5755a30	T4_chain_02	failure / malformed_arguments / 2	failure / correct_process_with_wrong_answer / 2	Steps 0-1 obtain revenue of 45000 and the correct 18% tax rate. Step 2 calculate('(20000 + 25000) * (1 - 0.18)') returns 36900 instead of the post-tax total 53100.	failure / correct_process_with_wrong_answer / 2
2	db85cc0b	T2_calc_02	failure / wrong_tool / 2	failure / premature_stop / 2	Step 1 calculate('40 * 150 + 15 * 150 * (1 + 0.0)') returns 8250, the correct answer. Steps 2-8 make several unproductive retrieval and file calls; step 9 submits an empty answer.	failure / loop_or_repetition / 2
2	ef115619	T2_calc_05	failure / loop_or_repetition / 0	failure / malformed_arguments / 0	Steps 0-4 call calculate with multiline code and variable assignments; each call is rejected with a syntax error. Step 5 submits an empty answer.	failure / malformed_arguments / 0
2	f7597e3e	T5_audit_02	failure / hallucinated_result / 2	failure / correct_process_with_wrong_answer / 2	Steps 0-1 read the payment log and SLA threshold of 350 ms. Four requests exceed 350 ms. Step 2 submits 1 without correctly counting all qualifying requests.	failure / correct_process_with_wrong_answer / 2

Appendix D: Supporting files

Evidence	Location
Raw Label Studio exports (unchanged), by round and annotator	`supporting_files/labels/`, `supporting_files/labels/round2/`
Agreement calculations, confusion matrices, disagreements	`supporting_files/agreement/round1/`, `round2/`, `pre_post_comparison.*`
Validation outputs on raw, processed, corrupted and augmented data	`supporting_files/validation/`
Grader v1 vs v2	`supporting_files/grader_v2/`, `data/processed/regrade_v2.csv`
Augmentation checks, examples, rejected candidates, spot check	`supporting_files/augmentation/`
Active-learning curves, plot data, selection manifests	`supporting_files/active_learning/`, `data/active_learning/`
Detector evaluation	`supporting_files/detectors/`
Split leakage report	`supporting_files/splits/leakage_report.json`
Dataset statistics; report traceability	`supporting_files/dataset_stats.json`, `supporting_files/report_traceability.csv`
Test log	`supporting_files/logs/`
2-minute screen recording of the pipeline	`supporting_files/recording/` (**to add**)